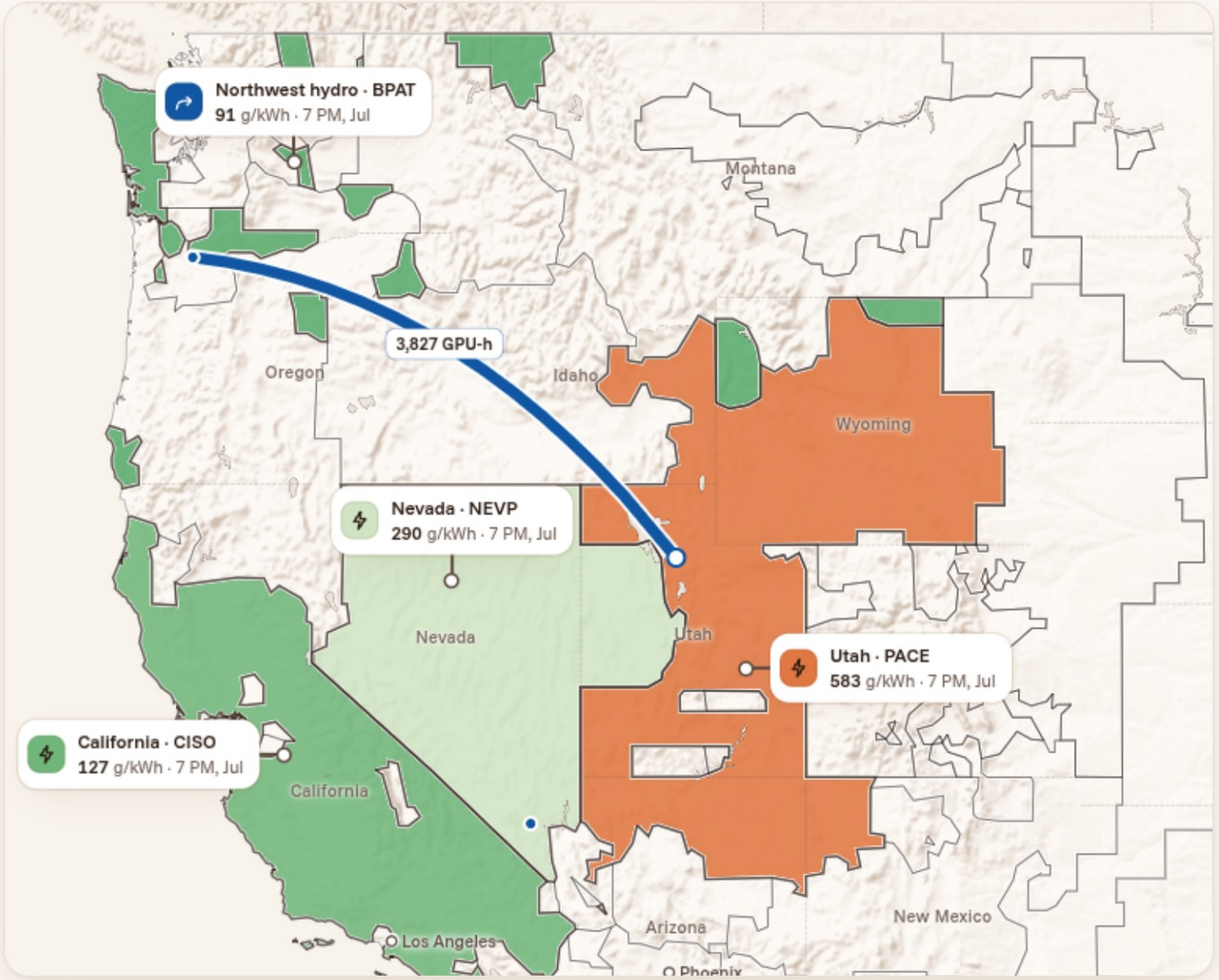
GridPilot: compute follows cleaner power

When and where should a Utah AI job run? We measured which lever matters.

A day-ahead traffic report for the grid, plus a router for AI jobs, tested in a simulated replay on real Utah grid data and a real GPU-cluster trace.



The Control room, test year: the router's work leaving Utah for the Northwest in one replay week.

Hard Tech Week 2026, Torus bounty: Energy × AI infrastructure

Open in dashboard: Story, step 1

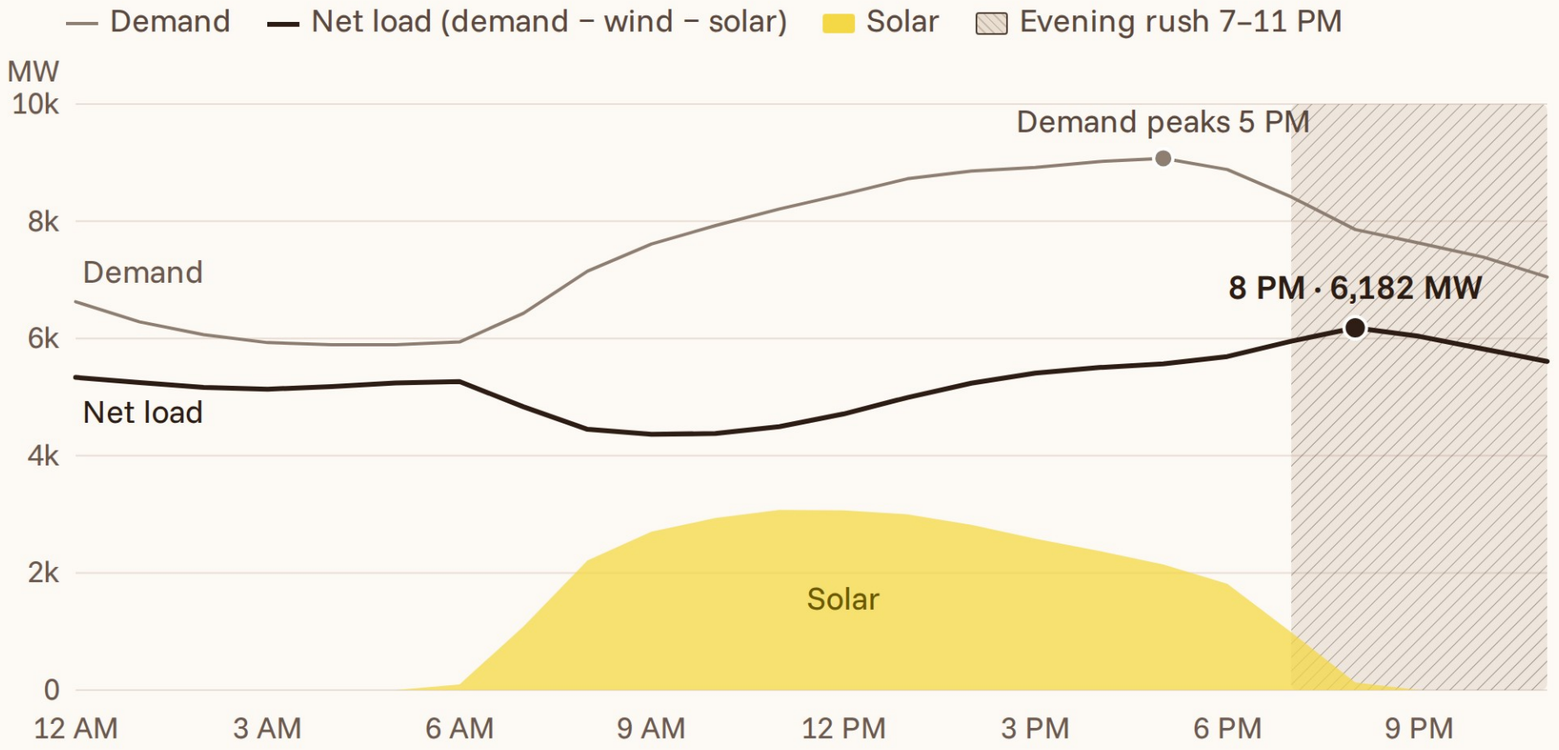
Utah's grid has a rush hour

As solar fades, Utah's net load peaks in the evening; the clean hours aren't the rush hours.

6,182 MW

net-load peak at 8 PM, mean of July 2026 weekdays in PACE, the balancing area serving much of Utah. Demand itself peaks at 5 PM.

Validation year: in 9 of 10 US regions we profiled, the rush hours are dirtier than average; Bonneville's are cleaner (−8.0%).



Evening rush: the hours holding at least 10 of the test year's 100 highest net-load hours, as in the dashboard's rush strip.

test year (sealed, opened once, 358 days) | observed · EIA-930

Open in dashboard: Story, step 2

One job: stay, wait or move

For a big training job, waiting buys almost nothing; moving changes everything.

Job 5930563#c5 from a public GPU-cluster trace, replayed onto Utah: 72 GPUs, 104 hours of pre-training.

Stay

Utah, PACE

2,098 kg CO_2

Runs as booked, on Utah's grid.

Wait

Utah, up to 6 hours late

2,098 kg CO_2

The wait-only policies kept its booked start: 6 hours of allowed delay buys almost nothing against a 104-hour run.

Move

the Northwest, Bonneville hydro

362 kg CO_2

30 minutes of transfer, then a hydro-heavy grid. Placed before it starts; nothing migrates mid-run.

test year (sealed, opened once, 358 days) | option CO_2 modeled | placements simulated | transfer and delay assumed

How it works

Forecast tomorrow, route each job, replay the year, count every kilowatt-hour.

1

Data

EIA-930 hourly grid data and EIA hourly CO_2 estimates, plus the Acme GPU-cluster trace.

2

10 AM day-ahead forecast

Only data published under assumed EIA delays: 2 h demand, 48 h wind and solar, 96 h carbon.

3

The router

Stay, wait or move. Every site has finite spare GPUs; every move pays a transfer delay and energy.

4

Exact replay

A gang-job simulator runs every policy on identical work, Utah's slot kept as each job's fallback.

5

Labelled results

Every number is observed, measured, modeled, forecast, assumed or simulated, and traces to a run.

Sealed test year: 358 days, opened once after a tagged freeze; 69 logged reads. 888,224 jobs replayed.

> AI agents helped build and review it. We froze the configuration before opening the holdout once; the routing policy remains a fixed ranking.

simulated replay | attributed, not avoided

Open in dashboard: Story, step 3

Moving beats waiting

22.3%

less attributed CO_2 on the sealed test year: policy S, the fixed clean-region ranking, against every job staying in Utah.

Its own 95% interval **17.3–27.2%**

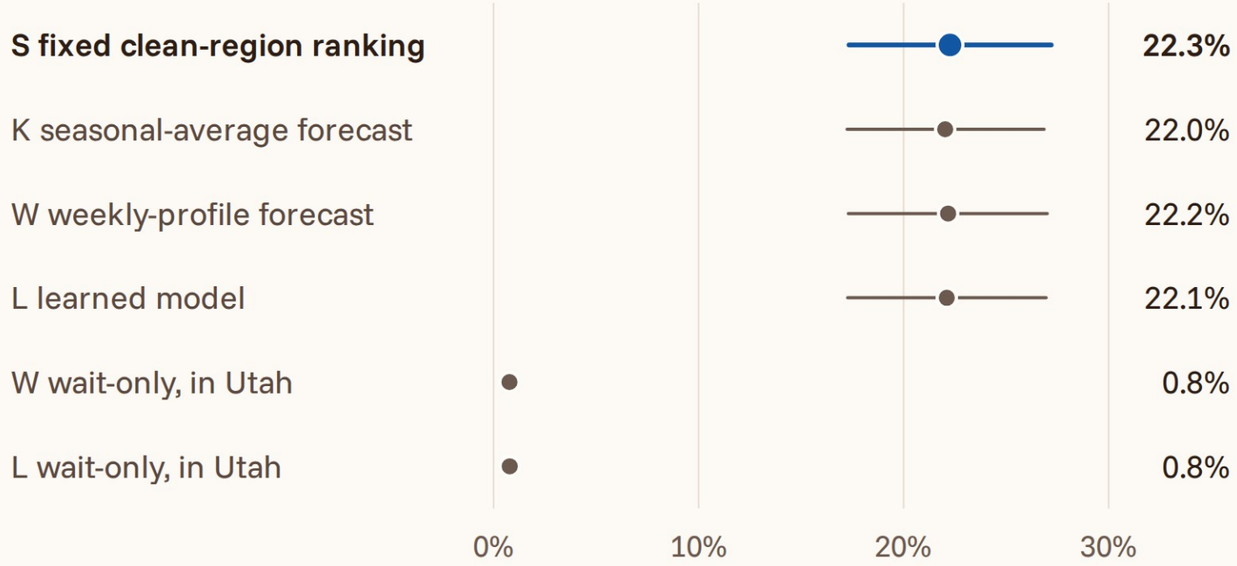
validation 22.6%

If only jobs of ≤ 8 GPUs may move:

3.2% [1.9, 4.7] test year; validation 3.2%

Saving vs every job staying in Utah, each policy's own 95% interval

Test year, 4-week blocks. Waiting in Utah alone: under 1% (0.8%). 70% of the work S ran outside Utah ran in the Northwest.



Simulated replay · 358 days · attributed (EIA consumption-basis) emissions, not avoided · deadlines met by construction (baseline bookings) · exact declared runtimes; runtime uncertainty untested.

test year (sealed, opened once, 358 days) | attributed, not avoided | simulated | baseline to pilot

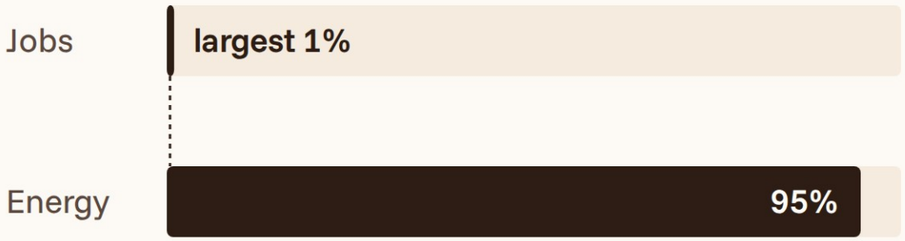
Open in dashboard: Story, step 4

The condition: can the big jobs move?

The saving lives in a few large jobs. If they can't move, most of it goes.

The largest 1% of jobs use 95% of the energy

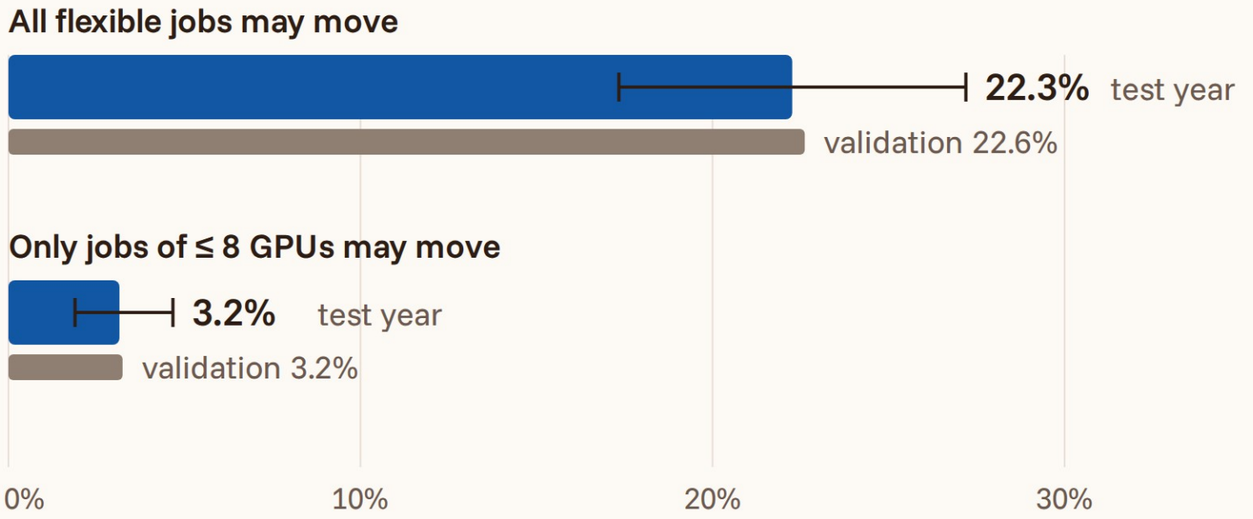
Test year, jobs ranked by declared GPU-hours (validation 94%).



Other changes move the result much less: a two-hour transfer still gives about 20%; twice the spare capacity, 35%. Both years agree.

S's saving vs every job staying in Utah, by which jobs may move

Blue: test year with its 95% interval; grey: validation. The forecast policies, which can also wait, reach 3.9% when only small jobs move.



test year (sealed, opened once, 358 days) | mobility assumed, not measured | scenario estimates

Open in dashboard: Story, step 5

Move what you can, store for what you can't

Aim the battery at carbon or at Utah's tightest hours, not both. An illustrative 4 MWh / 1 MW battery at the Utah site, on two fixed schedules registered before the test year was opened.

Test year, every job in Utah	**Aimed at carbon** the headline schedule	**Aimed at the tightest hours** peak-aimed schedule (D-073)
Cut in the cluster's attributed CO_2	**4.0%**	**1.0%**
On a flat marginal basis, storage losses	add about **107 t** CO_2	add about **77 t** CO_2
Average draw in Utah's 14 tightest hours (net load ≥ 7,207 MW)	0.88 → **0.55 MW**  6 of 14 hours discharging	0.88 → **0.32 MW**  10 of 14 hours discharging
The site's yearly peak draw	1.64 → 1.65 MW: charging fills the connection	1.64 → 1.65 MW: the same

Moving the router's work also adds 0.20 MW in Bonneville and 0.04 MW in California during Utah's tightest hours.

test year (sealed, opened once, 358 days) · illustrative hourly model, dispatch simulated · battery and η 0.85 assumed

Open in dashboard: Control room

What an operator does tomorrow

First identify what work can move. Then size storage for the Utah load that can't, aimed at carbon or at the rush.

1

Tag movable jobs

Measure which real jobs can move: checkpointing, data size, residency. Mobility decides the result.

2

Pilot the fixed ranking

Policy S is the baseline to pilot: 22.3% on the test year, 3.2% if only small jobs move.

3

Size storage for the rest

For the residual Utah load: 0.83 MW in the tightest hours if only ≤ 8-GPU jobs move (forecast policy W).

Limits Simulated replay; mobility assumed; no reservation-free or runtime-overrun test. Cost only as a wholesale scenario, validation year: moving costs +7–9% vs every job in Utah, about $21 per attributed tonne.

Next Measure which real jobs can move, with an operator; couple the router's residual Utah load to storage dispatch; with a storage partner, test seconds-scale buffering on measured cluster power.

test year (sealed, opened once, 358 days) | simulated replay | attributed, not avoided

Appendix, for Q&A

Does it hold? Validation beside the sealed test year

S vs every job in Utah, 4-week blocks	Validation	Test (sealed)
Registered primary	**22.6%**	**22.3%**
Workload phase 49	23.2%	23.1%
Workload phase 98	22.0%	23.3%
Two-hour transfer	20.2%	19.9%
24-hour batch delay	22.8%	22.3%
No spare destination slots	0.0%	0.0%
Twice the spare capacity (1,024)	35.0%	34.9%
Only jobs of ≤ 8 GPUs may move	3.2%	3.2%
Production-basis repricing	25.5%	validation only
Observed hours only (common support)	22.5%	validation only

Rows 2–8: one-axis scenarios, pre-registered secondary runs on both years (D-071). The last two rows are validation only. Scenario estimates, not a confidence interval.

simulated | scenario estimates | results/test/extras.json drift table; results/extensions.json

Appendix, for Q&A

Water, illustrative: no water-saving claim

S vs every job in Utah, generation-associated water	Validation	Test (sealed)
Without hydro evaporation	−20.5%	−18.6%
With hydro evaporation	+145%	+174%

- Moving to the Northwest's hydro trades thermal-plant water for reservoir evaporation; the sign depends on whether evaporation counts.
- Consumption is not withdrawal; production-side weighting; PACE is not the Great Salt Lake watershed.
- Factors: NREL/TP-6A20-50900 median consumption; direct cooling from the LBNL 2024 WUE range.

simulated illustrative

Appendix, for Q&A

The claim frame and the limitations

Claim frame (verbatim): “simulated advance-reservation scheduling scenario; baseline-relative deadlines; exact declared runtimes; runtime uncertainty untested; EIA-attributed (consumption-basis) emissions; conditional temporal block-resampling intervals”

- Publication-lagged replay of a revised historical snapshot; real-time vintages are not reconstructed. Publication delays are assumed availability rules.
- EIA's consumption-basis intensity assigns imports from Canada and Mexico the average US emissions rate.
- Zero deadline misses hold by construction under baseline bookings; not evidence of production reliability. The bookings' effect on realized savings is not signed.
- Shadow-U ("possible without advance reservations"): cut (D-058).* Phases 49 and 98: run as pre-registered secondary scenarios.*
- The test year was opened once, through the logged entry point, after the freeze; nothing was selected or refitted on it. It covers the days up to the snapshot's cutoff, not a full twelve months.
- EIA's 2025–26 emission factors may be provisional. Baseline bookings are an assumed advance-reservation calendar, not a historical online-router claim. Missing truth hours are filled by pre-registered tiers.

* Errata: the frozen results/test/manifest.json still reads "pending" for these two lines, written before the extras ran (audit F16). The results file is not edited.

claim frame verbatim from results/test/manifest.json

Appendix, for Q&A

The seal: one opening, logged

- One opening at 19:22 MT on Tue Oct 27 (2026-10-28T01:22:53Z), after the tag **config-freeze**; 69 logged reads.
- Test year: 2025-11-01 to 2026-10-24 inclusive, 358 days, to the snapshot's last complete day.
- Excluded tail: arrivals after Oct 15, 23:04 UTC, 8,003 jobs (3.1% of GPU-hours), the same for every policy, so every deadline falls before the cutoff (validation: 1,203 jobs, 3.9%).
- Disclosed before the opening: counts-only completeness checks (D-052, D-074).
- A blocked early attempt (D-060).
- The column names and node list of one post-seal price file printed, with no values (D-072).
- An empty `data/.wtest` file created and deleted by a reviewer, changing only the directory's mtime (D-074).

results/test_reads.jsonl

config/freeze.json

Appendix, for Q&A

Data sources and licences

Source	Licence
EIA-930 hourly balance and EIA hourly CO_2 estimates	U.S. Government work, public domain
Acme GPU-cluster trace, Seren cluster	CC BY 4.0
Open-Meteo archived forecasts (toughest-days case study)	CC BY 4.0: Weather data by Open-Meteo.com
NREL and LBNL water factors	DOE national-laboratory reports, cited
CAISO/WEIM real-time LMPs (wholesale scenario, not a bill)	CAISO OASIS public market data
Electricity Maps balancing-area boundaries	AGPL-3.0
Natural Earth	Public domain
Funnel Sans and Funnel Display, embedded in this deck	SIL Open Font License 1.1

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